

Reservation FE

Find Flights:

From:

To:

Departure Date:

Available Flights:

Airlines	Departure City	Arrival City	EstimateTime	Departure	Action
Cambodia Airline	KTI	VN	2026-06-01T03:14:07		Select
Cambodia Airline	KTI	VN	2026-06-01T03:14:07		Select
Cambodia Airline	KTI	VN	2026-06-05T03:14:07		Select

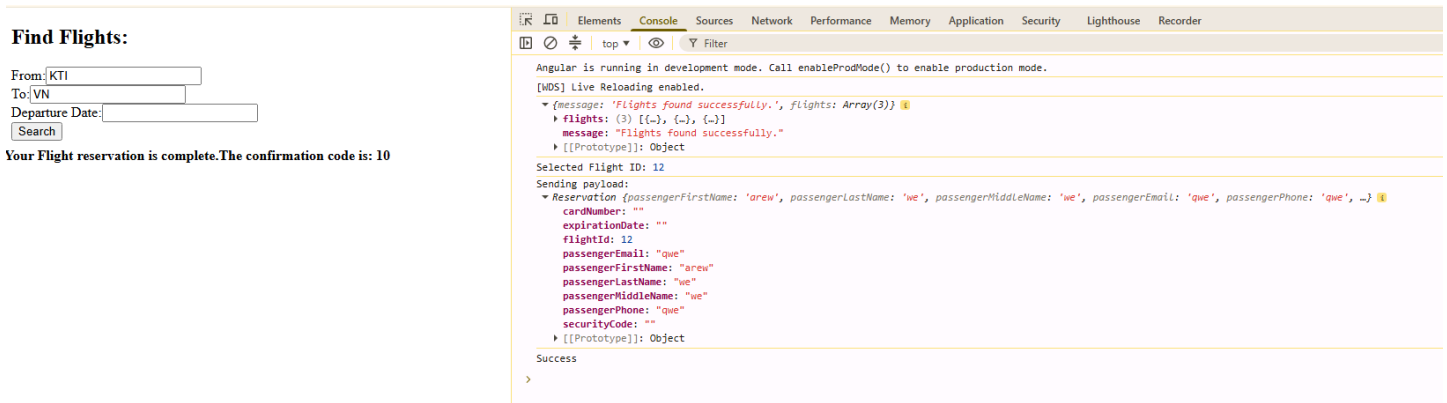
Complete Reservation:

Flight Details:

Airline: Cambodia Airline
Departure City: KTI
Arrival City: VN
Departure Date: 2026-06-01

Passenger Details:

First Name:		
Last Name:		
Middle Name:		
Email:		
Phone:		
Card Number:		
Expiry Date:		Security Code:



1-Create service Reservation

```
reservation.service.ts X
src > app > services > reservation.service.ts > ...
1  import { Injectable } from '@angular/core';
2  import { HttpClient } from '@angular/common/http';
3  import { Reservation } from '../model/reservation';
4  import { Observable } from 'rxjs';
5
6  @Injectable({
7    providedIn: 'root'
8  })
9  export class ReservationService {
10
11    // Moved to private to follow best practices
12    private reservationUrl: string = "http://localhost:8080/api/reservations";
13
14    constructor(private _httpClient: HttpClient) { }
15
16    // FIX: Return the Observable directly so your components can subscribe to it
17    public getAllReservation(): Observable<any> {
18      return this._httpClient.get<any>(this.reservationUrl);
19    }
20
21    // FIX: Return the Observable directly so your components can subscribe to it
22    public getReservationById(id: number): Observable<any> {
23      return this._httpClient.get<any>(`${this.reservationUrl}/${id}`);
24    }
25
26    // This one was already correct!
27    public saveReservation(reservation: Reservation): Observable<any> {
28      return this._httpClient.post<any>(this.reservationUrl, reservation);
29    }
30
31  }
```

- Import httpClient : សម្រាប់ Http Method GET, POST, PUT, Delete
- constructor(private _httpClient: HttpClient) { } ពេលបង្កើត object នឹងមកអោយវាហៅ Http មកប្រើប្រាស់
- បន្ទាប់មកយើងបង្កើត Method ដើម្បីធ្វើការប្រើប្រាស់ : getAllReservation(), and getReservationById(id: number), saveReservation(reservation: Reservation).

- Observable<any> យើង Return វាជា Observable ព្រោះយើងគិតវាដូចជា អ្នកបំផ្លិច វា អ្នកទៅយកពីខាងក្រោយ API
- សម្រាប់ subscribe យើងជាអ្នកប្រើប្រាស់

សំខាន់ដើម្បីប្រើប្រាស់ HttpClient បានយើងត្រូវ ដាក់ HttpClientModule to the app.module.ts

```

app.module.ts X
src > app > app.module.ts > AppModule
1 import { NgModule } from '@angular/core';
2 import { BrowserModule } from '@angular/platform-browser';
3
4 import { AppRoutingModule } from './app-routing.module';
5 import { AppComponent } from './app.component';
6 import { FindFlightsComponent } from './components/find-flights/find-flights.component';
7 import { DisplayFlightsComponent } from './components/display-flights/display-flights.component';
8 import { PassengerDetailsComponent } from './components/passenger-details/passenger-details.component';
9 import { ConfirmComponent } from './components/confirm/confirm.component';
10 import { FormsModule } from '@angular/forms';
11 import { HttpClientModule } from '@angular/common/http';
12 import { ReservationService } from './services/reservation.service';
13 import { FlightService } from './services/flight.service';
14
15 @NgModule({
16   declarations: [
17     AppComponent,
18     FindFlightsComponent,
19     DisplayFlightsComponent,
20     PassengerDetailsComponent,
21     ConfirmComponent
22   ],
23   imports: [
24     BrowserModule,
25     AppRoutingModule,
26     FormsModule,
27     HttpClientModule
28   ],
29   providers: [ReservationService, FlightService],
30   bootstrap: [FindFlightsComponent]
31 })
32 export class AppModule { }
33

```

- នឹងត្រូវដាក់ FormsModule ដើម្បីធ្វើការប្រើប្រាស់ form បាន

2- បង្កើត model

```

TS reservation.ts X
src > app > model > reservation.ts > Reservation
1 export class Reservation {
2   constructor(
3     public passengerFirstName: string,
4     public passengerLastName: string,
5     public passengerMiddleName: string,
6     public passengerEmail: string,
7     public passengerPhone: string,
8     public cardNumber: string,
9     public expirationDate: string,
10    public securityCode: string,
11    public flightId: number // Changed from string to number
12  ) { }
13 }

```

ពេលយើងចង់បង្កើត object from our model we use “new” keyword

- reservation:Reservation = new Reservation("", "", "", "", "", "", "", "", 0);

3- ហៅ Server Reservation មកប្រើប្រាស់

```
passenger-details.component.ts X
src > app > components > passenger-details > passenger-details.component.ts > PassengerDetailsComponent > ngOnInit
1  import { Component, OnInit } from '@angular/core';
2  import { ActivatedRoute, Router } from '@angular/router';
3  import { Reservation } from 'src/app/model/reservation';
4  import { FlightService } from 'src/app/services/flight.service';
5  import { ReservationService } from 'src/app/services/reservation.service';
6
7  @Component({
8    selector: 'app-passenger-details',
9    templateUrl: './passenger-details.component.html',
10   styleUrls: ['./passenger-details.component.css']
11 })
12 export class PassengerDetailsComponent implements OnInit {
13
14   flightData:any;
15
16   reservation:Reservation = new Reservation("", "", "", "", "", "", "", "", 0);
17
18   constructor(private _flightService: FlightService,
19     private _reservationService:ReservationService
20     ,
21     private _router: Router, private route:ActivatedRoute) { }
22
23   ngOnInit(): void {
24     const id = this.route.snapshot.paramMap.get("id");
25     if (id){
26       this._flightService.getFlight(Number.parseInt(id)).subscribe((res:any)=>{
27         this.flightData = res;
28       });
29     }
30
31   }
32
33   // Triggered when the form is submitted
34   onSubmit(): any {
35     // Parse to integer to guarantee it matches the backend 'int' type
36     this.reservation.flightId = Number.parseInt(this.flightData.id);
37
38     console.log("Sending payload:", this.reservation);
39
40     this._reservationService.saveReservation(this.reservation).subscribe((res:any)=>{
41       console.log("Success");
42       this._router.navigate(['/confirm/' + res.id]);
43     });
44
45   }
46
47 }
48
```

- ហៅ Service មកប្រើប្រាស់ មុនគេ: ActivatedRoute ប្រើប្រាស់ដើម្បីធ្វើការចាប់យក ទិន្នន័យពី url ដែលបានបោះមក (<http://localhost:4200/confirm/10>) យើងចង់បាន id លេខ 10 នឹងយើងត្រូវប្រើប្រាស់ `const id = this.route.snapshot.paramMap.get("id");`

```
constructor(private _flightService: FlightService,
  private _reservationService: ReservationService
  ,
  private _router: Router, private route: ActivatedRoute) { }
```

- `onSubmit()`: គឺជា function ដែលយើងធ្វើការ submit form from temple

```
16 <h2>Passenger Details:</h2>
17 <form #passengerForm="ngForm" (ngSubmit)="onSubmit()">
18   <pre>
19   First Name: <input type="text" name="passengerFirstName" [(ngModel)]="reservation.passengerFirstName"/>
20   Last Name: <input type="text" name="passengerLastName" [(ngModel)]="reservation.passengerLastName"/>
21   Middle Name: <input type="text" name="passengerMiddleName" [(ngModel)]="reservation.passengerMiddleName"/>
22   Email: <input type="text" name="passengerEmail" [(ngModel)]="reservation.passengerEmail"/>
23   Phone: <input type="text" name="passengerPhone" [(ngModel)]="reservation.passengerPhone"/>
24   Card Number: <input type="text" name="cardNumber" [(ngModel)]="reservation.cardNumber"/>
25   Expiry Date: <input type="text" name="expirationDate" [(ngModel)]="reservation.expirationDate"/>
26   Security Code: <input type="text" name="securityCode" [(ngModel)]="reservation.securityCode"/>
27
28   <input type="submit" value="Confirm Reservation"/>
29 </pre>
30 </form>
```

- `name="passengerFirstName" [(ngModel)]`: គេធ្វើ two way property binding ដែលវាធ្វើការភ្ជាប់ ពី Template to the ts variable.

4- Router

- យើងធ្វើការបង្កើតដើម្បីប្រាប់ URL ថាវាទៅណា { path: "", redirectTo: "", pathMatch: 'full' },

ALL Code in FE:

```
index.html X
src > index.html > ...
1  <!doctype html>
2  <html lang="en">
3  <head>
4    <meta charset="utf-8">
5    <title>FlightReservation</title>
6    <base href="/">
7    <meta name="viewport" content="width=device-width, initial-scale=1">
8    <link rel="icon" type="image/x-icon" href="favicon.ico">
9  </head>
10 <body>
11   <app-find-flights></app-find-flights>
12 </body>
13 </html>
14
```

```
app.module.ts X
src > app > app.module.ts > AppModule
1  import { NgModule } from '@angular/core';
2  import { BrowserModule } from '@angular/platform-browser';
3
4  import { AppRoutingModule } from './app-routing.module';
5  import { AppComponent } from './app.component';
6  import { FindFlightsComponent } from './components/find-flights/find-flights.component';
7  import { DisplayFlightsComponent } from './components/display-flights/display-flights.component';
8  import { PassengerDetailsComponent } from './components/passenger-details/passenger-details.component';
9  import { ConfirmComponent } from './components/confirm/confirm.component';
10 import { FormsModule } from '@angular/forms';
11 import { HttpClientModule } from '@angular/common/http';
12 import { ReservationService } from './services/reservation.service';
13 import { FlightService } from './services/flight.service';
14
15 @NgModule({
16   declarations: [
17     AppComponent,
18     FindFlightsComponent,
19     DisplayFlightsComponent,
20     PassengerDetailsComponent,
21     ConfirmComponent
22   ],
23   imports: [
24     BrowserModule,
25     AppRoutingModule,
26     FormsModule,
27     HttpClientModule
28   ],
29   providers: [ReservationService, FlightService],
30   bootstrap: [FindFlightsComponent]
31 })
32 export class AppModule { }
33
```

```

app-routing.module.ts X
src > app > app-routing.module.ts > ...
1  import { NgModule } from '@angular/core';
2  import { RouterModule, Routes } from '@angular/router';
3  import { FindFlightsComponent } from '../components/find-flights/find-flights.component';
4  import { DisplayFlightsComponent } from '../components/display-flights/display-flights.component';
5  import { PassengerDetailsComponent } from '../components/passenger-details/passenger-details.component';
6  import { ConfirmComponent } from '../components/confirm/confirm.component';
7
8  const routes: Routes = [
9    { path: '', redirectTo: '', pathMatch: 'full' },
10   {
11     path: 'findFlight',
12     component: FindFlightsComponent,
13   },
14
15   {
16     path: 'displayFlights',
17     component: DisplayFlightsComponent,
18   },
19   {
20     path: 'passengerDetails/:id',
21     component: PassengerDetailsComponent,
22   },
23   {
24     path: 'confirm/:id',
25     component: ConfirmComponent,
26   },
27 ];
28
29 @NgModule({
30   imports: [RouterModule.forRoot(routes)],
31   exports: [RouterModule],
32 })
33 export class AppRoutingModule {}
34

```

```

reservation.service.ts X
src > app > services > reservation.service.ts > ReservationService
1  import { Injectable } from '@angular/core';
2  import { HttpClient } from '@angular/common/http';
3  import { Reservation } from '../model/reservation';
4  import { Observable } from 'rxjs';
5
6  @Injectable({
7    providedIn: 'root'
8  })
9  export class ReservationService {
10
11    // Moved to private to follow best practices
12    private reservationUrl: string = "http://localhost:8080/api/reservations";
13
14    constructor(private _httpClient: HttpClient) { }
15
16    // FIX: Return the Observable directly so your components can subscribe to it
17    public getAllReservation(): Observable<any> {
18      return this._httpClient.get<any>(this.reservationUrl);
19    }
20
21    // FIX: Return the Observable directly so your components can subscribe to it
22    public getReservationById(id: number): Observable<any> {
23      return this._httpClient.get<any>(`${this.reservationUrl}/${id}`);
24    }
25
26    // This one was already correct!
27    public saveReservation(reservation: Reservation): Observable<any> {
28      return this._httpClient.post<any>(this.reservationUrl, reservation);
29    }
30
31  }

```



```

flight.service.ts X
src > app > services > flight.service.ts > FlightService > getFlight
1  import { Injectable } from '@angular/core';
2  import { HttpClient } from '@angular/common/http';
3  import { Observable } from 'rxjs';
4  import { Criteria } from '../model/criteria';
5
6  @Injectable({
7    providedIn: 'root'
8  })
9  export class FlightService {
10
11    flightUrl: string = "http://localhost:8080/api/flights/search";
12
13    // This will store the raw flights array for DisplayFlightsComponent
14    data: any[] = [];
15
16    constructor(private _httpClient: HttpClient) { }
17
18    // 1. Returns the Observable directly so FindFlightsComponent can subscribe to it
19    public getFlights(criteria: Criteria): Observable<any> {
20      const url = `${this.flightUrl}?from=${criteria.from}&to=${criteria.to}&departureDate=${criteria.departureDate}`;
21      return this._httpClient.get<any>(url);
22    }
23
24    // 2. Returns the Observable for a single flight detail lookup
25    public getFlight(id: number): Observable<any> {
26      return this._httpClient.get<any>("http://localhost:8080/api/flights/" + id);
27    }
28
29  }

```

- Model

```

TS reservation.ts X
src > app > model > TS reservation.ts > Reservation
1  export class Reservation {
2    constructor(
3      public passengerFirstName: string,
4      public passengerLastName: string,
5      public passengerMiddleName: string,
6      public passengerEmail: string,
7      public passengerPhone: string,
8      public cardNumber: string,
9      public expirationDate: string,
10     public securityCode: string,
11     public flightId: number // Changed from string to number
12   ) { }
13 }

```

```

TS criteria.ts X
src > app > model > TS criteria.ts > Criteria
1  export class Criteria {
2
3    constructor(
4      public from:string,
5      public to:string,
6      public departureDate:string) { }
7
8  }

```


- Display Component

```
display-flights.component.ts X
src > app > components > display-flights > display-flights.component.ts > DisplayFlightsComponent
1  import { Component, OnInit } from '@angular/core';
2  import { Route, Router } from '@angular/router';
3  import { FlightService } from 'src/app/services/flight.service';
4
5  @Component({
6    selector: 'app-display-flights',
7    templateUrl: './display-flights.component.html',
8    styleUrls: ['./display-flights.component.css']
9  })
10 export class DisplayFlightsComponent implements OnInit {
11
12    data:any;
13
14    constructor(private _flightService: FlightService, private _router: Router) { }
15
16    ngOnInit(): void {
17      this.data = this._flightService.data;
18    }
19
20    public onSelect(id: number): void {
21      console.log('Selected Flight ID:', id);
22      this._router.navigate(['/passengerDetails/'+id])
23    }
24 }
```

```
display-flights.component.html X
src > app > components > display-flights > display-flights.component.html > html > body >
1  <html>
2    <head>
3      <title>Flights</title>
4    </head>
5    <body>
6      <h2>Available Flights:</h2>
7
8      <table>
9        <thead>
10         <tr>
11           <th>Airlines</th>
12           <th>Departure City</th>
13           <th>Arrival City</th>
14           <th>EstimateTimeDeparture</th>
15           <th>Action</th>
16         </tr>
17       </thead>
18       <tbody>
19         <tr *ngFor="let flight of data">
20           <td>{{ flight.operatingAirlines }}</td>
21           <td>{{ flight.departureCity }}</td>
22           <td>{{ flight.arrivalCity }}</td>
23           <td>{{ flight.estimatedDepartureTime }}</td>
24           <td>
25             <a href="javascript:void(0)" (click)="onSelect(flight.id)"
26               >Select</a>
27           </td>
28         </tr>
29       </tbody>
30     </table>
31   </body>
32 </html>
```

- Find-flight component

```
find-flights.component.ts X
src > app > components > find-flights > find-flights.component.ts > ...
1  import { Component, OnInit } from '@angular/core';
2  import { Router } from '@angular/router';
3  import { Criteria } from 'src/app/model/criteria';
4  import { FlightService } from 'src/app/services/flight.service';
5
6  @Component({
7    selector: 'app-find-flights',
8    templateUrl: './find-flights.component.html',
9    styleUrls: ['./find-flights.component.css']
10 })
11 export class FindFlightsComponent implements OnInit {
12
13   criteria: Criteria = new Criteria('', '', '');
14
15   constructor(private _flightService: FlightService, private _router: Router) { }
16
17   ngOnInit(): void {
18   }
19
20   public onSubmit(): void {
21     this._flightService.getFlights(this.criteria).subscribe(
22       (res: any) => {
23         // Assign the array of flights from your API payload to the service data
24         this._flightService.data = res.flights;
25         console.log(res)
26         // Navigate to the list component
27         this._router.navigate(['/displayFlights']);
28       },
29       (error) => {
30         console.error("Error fetching flights", error);
31       }
32     );
33   }
34
35 }
```

```
find-flights.component.html X
src > app > components > find-flights > find-flights.component.html > html > body > fo
1  <html>
2    <head>
3      <title>Find Flights</title>
4    </head>
5    <body>
6      <h2>Find Flights:</h2>
7
8      <form #findFlightsForm="ngForm" (ngSubmit)="onSubmit()">
9        <div>
10         <label>From:</label>
11         <input type="text" [(ngModel)]="criteria.from" name="from" />
12       </div>
13
14       <div>
15         <label>To:</label>
16         <input type="text" [(ngModel)]="criteria.to" name="to" />
17       </div>
18
19       <div>
20         <label>Departure Date:</label>
21         <input
22           type="text"
23           [(ngModel)]="criteria.departureDate"
24           name="departureDate"
25         />
26       </div>
27
28       <button type="submit">Search</button>
29     </form>
30   </body>
31 </html>
32 <router-outlet></router-outlet>
33
```

- Passenger-detail

```
import { Component, OnInit } from '@angular/core';
import { ActivatedRoute, Router } from '@angular/router';
import { Reservation } from 'src/app/model/reservation';
import { FlightService } from 'src/app/services/flight.service';
import { ReservationService } from 'src/app/services/reservation.service';
```

```
@Component({
  selector: 'app-passenger-details',
  templateUrl: './passenger-details.component.html',
  styleUrls: ['./passenger-details.component.css']
})
export class PassengerDetailsComponent implements OnInit {
```

```
  flightData:any;
```

```
  reservation:Reservation = new Reservation("", "", "", "", "", "", "", 0);
```

```
  constructor(private _flightService: FlightService,
    private _reservationService:ReservationService
    ,
    private _router: Router, private route:ActivatedRoute) { }
```

```
  ngOnInit(): void {
    const id = this.route.snapshot.paramMap.get("id");
    if (id){
      this._flightService.getFlight(Number.parseInt(id)).subscribe((res:any)=>{
        this.flightData = res;
      });
    }
  }
```

```
  // Triggered when the form is submitted
```

```
  onSubmit(): any {
    // Parse to integer to guarantee it matches the backend 'int' type
    this.reservation.flightId = Number.parseInt(this.flightData.id);
```

```
    console.log("Sending payload:", this.reservation);
```

```
    this._reservationService.saveReservation(this.reservation).subscribe((res:any)=>{
      console.log("Success");
      this._router.navigate(['/confirm/' + res.id]);
    });
```

```
  }
```

```
}
```

```
<html>
<head>
  <title>Complete Reservation</title>
</head>
<body>
  <h2>Complete Reservation:</h2>
```

```

  <h2>Flight Details:</h2>
  <p>
    <strong>Airline:</strong> {{ flightData?.operatingAirlines }}<br />
    <strong>Departure City:</strong> {{ flightData?.departureCity }}<br />
    <strong>Arrival City:</strong> {{ flightData?.arrivalCity }}<br />
    <strong>Departure Date:</strong> {{ flightData?.dateOfDeparture }}<br />
  </p>
```

```
  <h2>Passenger Details:</h2>
  <form #passengerForm="ngForm" (ngSubmit)="onSubmit()">
    <pre>
First Name:  <input type="text" name="passengerFirstName" [(ngModel)]="reservation.passengerFirstName"/>
Last Name:   <input type="text" name="passengerLastName" [(ngModel)]="reservation.passengerLastName"/>
Middle Name: <input type="text" name="passengerMiddleName" [(ngModel)]="reservation.passengerMiddleName"/>
Email:       <input type="text" name="passengerEmail" [(ngModel)]="reservation.passengerEmail"/>
Phone:       <input type="text" name="passengerPhone" [(ngModel)]="reservation.passengerPhone"/>
Card Number: <input type="text" name="cardNumber" [(ngModel)]="reservation.cardNumber"/>
Expiry Date: <input type="text" name="expirationDate" [(ngModel)]="reservation.expirationDate"/>
Security Code: <input type="text" name="securityCode" [(ngModel)]="reservation.securityCode"/>
```

```
    <input type="submit" value="Confirm Reservation"/>
```

```
  </pre>
</form>
```

```
</body>
</html>
```

- Confirm-detail

```
confirm.component.ts X
src > app > components > confirm > confirm.component.ts > ...
1  import { Component, OnInit } from '@angular/core';
2  import { ActivatedRoute, Router } from '@angular/router';
3  import { FlightService } from 'src/app/services/flight.service';
4  import { ReservationService } from 'src/app/services/reservation.service';
5
6
7  @Component({
8    selector: 'app-confirm',
9    templateUrl: './confirm.component.html',
10   styleUrls: ['./confirm.component.css']
11 })
12 export class ConfirmComponent implements OnInit {
13
14   reservationId!: number;
15
16   constructor(private _flightService: FlightService,
17     private _reservationService: ReservationService
18   ,
19     private _router: Router, private route: ActivatedRoute) { }
20
21   ngOnInit(): void {
22
23     this.reservationId = Number.parseInt(this.route.snapshot.paramMap.get("id")!);
24   }
25
26 }
```

Your Flight reservation is complete.The confirmation code is:
{{reservationId}}